

Robustness or Weighting? Re-Evaluating Federated Aggregation on Heterogeneous Health Data

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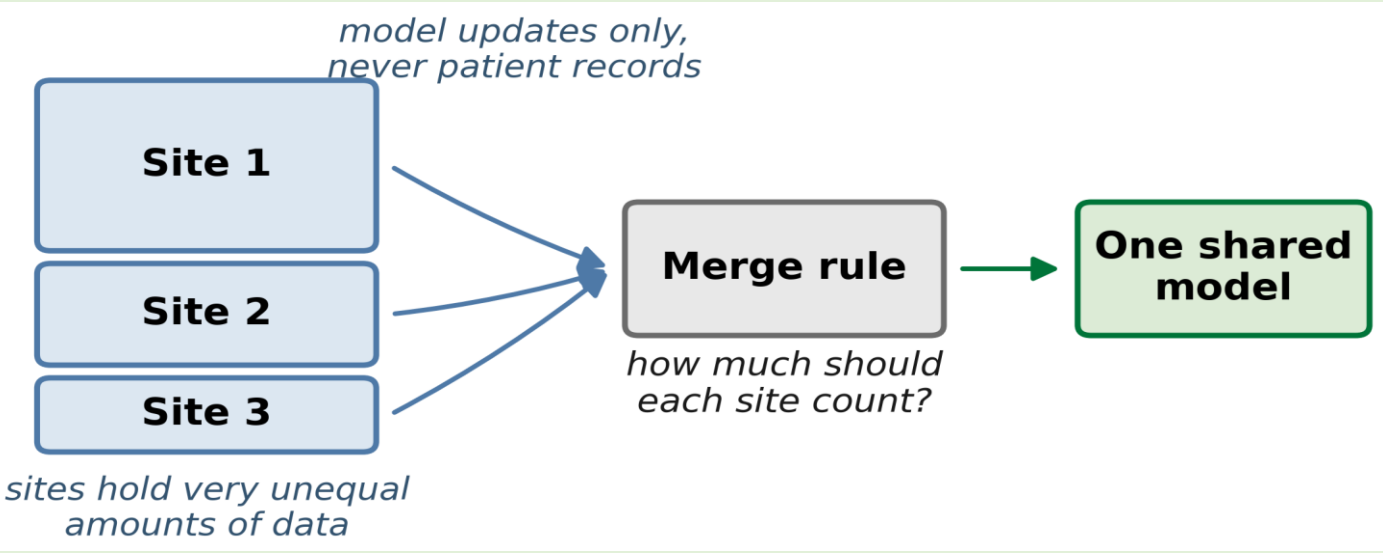


Motivation & Questions

- Hospitals and health agencies hold records that could sharpen how we predict disease, but privacy law stops them pooling the data.
- Federated learning trains one shared model without moving any patient record out of the institution holding it.
- The rule that merges each site's work is normally chosen for its robustness to noisy or unusual sites.
- Robust rules are always scored against FedAvg, which weights each site by how much data it holds, so the biggest site gets the loudest vote.
- Robust rules discard that weighting entirely, so two things change at once and nobody had separated them.
- We ask: when a robust rule wins, is it the robust statistic winning, or just the end of the loudest vote?

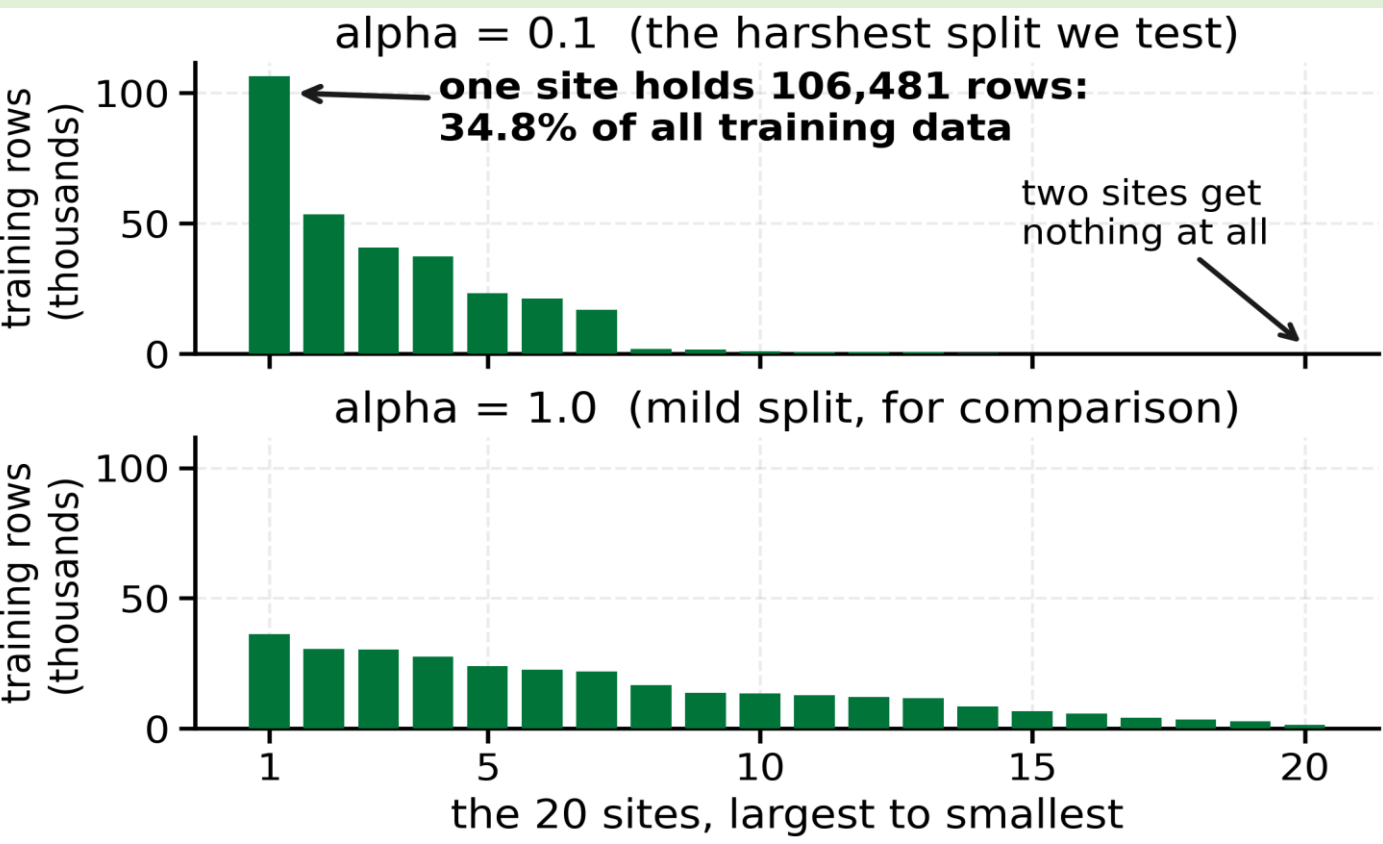
Background: how federated learning works

- Each site trains on its own data and sends only the model update, which the server merges into one shared model.



Dataset: two health datasets, split 20 ways

- CDC BRFSS 2023: 382,709 U.S. adults, predicting self-reported depression, which 20.7% of respondents report.
- UCI Breast Cancer Wisconsin runs alongside as a clean clinical benchmark.
- Each dataset is dealt to 20 simulated sites by Dirichlet partitioning, whose dial alpha sets how lopsided the split is.



- At alpha = 0.1 one site holds 106,481 training rows, 34.8% of the total, while two sites receive nothing at all.

Methods: seven merge rules and one control

- Seven merge rules run head to head: FedAvg, FedProx, CS-Agg, Krum, trimmed mean, coordinate-wise median, and an unweighted mean.
- The unweighted mean is our control: it uses no robust statistic and differs from FedAvg only in ignoring how much data each site holds.
- Anything the control gains over FedAvg is therefore weighting alone, which is what isolates the effect.

References

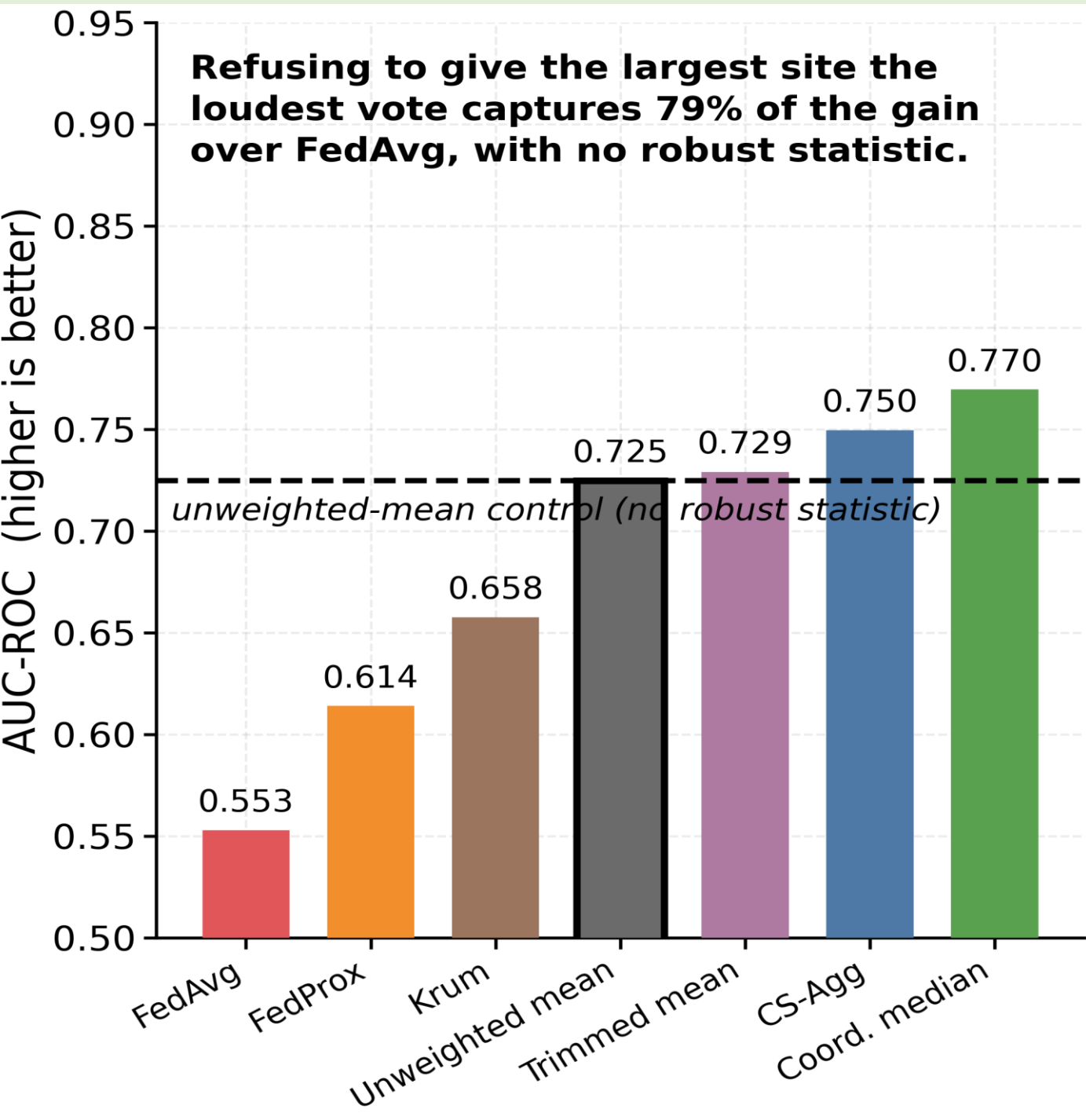
[1] McMahan et al. Communication-efficient learning of deep networks from decentralized data. AISTATS, 2017.
[2] Blanchard et al. Machine learning with adversaries: Byzantine tolerant gradient descent. NIPS, 2017.
[3] Yin et al. Byzantine-robust distributed learning: towards optimal statistical rates. ICML, 2018.
[4] Karimireddy et al. Byzantine-robust learning on heterogeneous datasets via bucketing. ICLR, 2022.
[5] Hsu et al. Measuring the effects of non-identical data distribution for federated visual classification. arXiv:1909.06335, 2019.
[6] CDC. BRFSS 2023 survey data and documentation.

Methods, continued

- Labels are flipped at 20% of sites, across 0 to 30% of their records.
- On BRFSS we flip them only the way real people err, reporting no depression when they have it.
- 624 conditions cover every combination of rule, dataset, split, noise level and random seed.
- Each runs 50 rounds, scored by AUC-ROC on a clean held-out test set the sites never see.
- Differences are tested with Wilcoxon signed-rank, Holm-corrected within each of three comparison families.

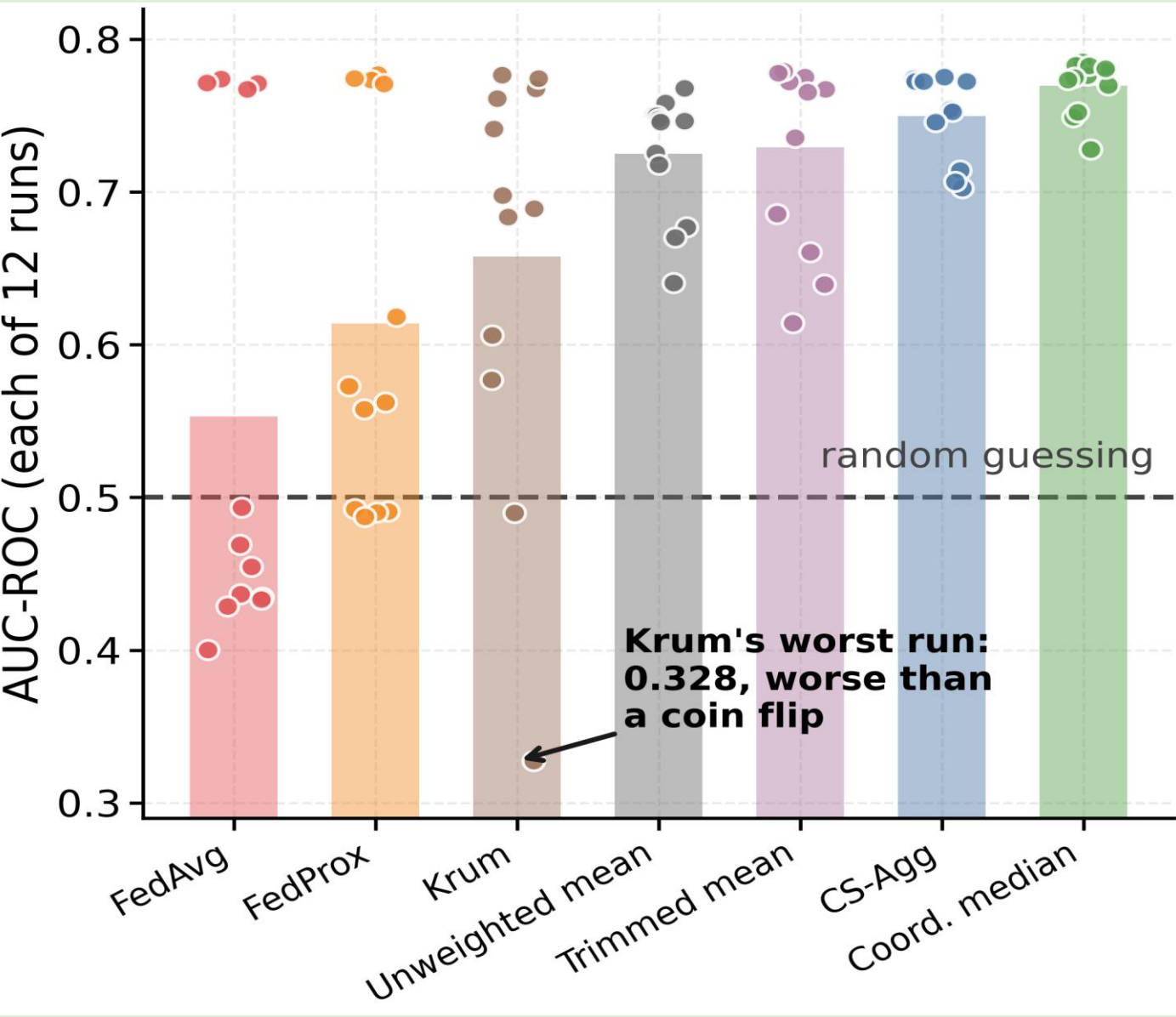
Results

Most of the gain over FedAvg is weighting, not robustness



- The control alone recovers 79 to 98% of what the three strongest robust rules gain over FedAvg.
- Only coordinate-wise median clearly clears it, at +0.0163 AUC (Holm $p < 0.0001$).
- Trimmed mean's +0.0026 is significant but negligible, and CS-Agg cannot be told apart from the control ($p = 0.22$).

Krum is unreliable, not merely weaker



- Krum lands below the control and collapses to 0.328 AUC on one run, worse than a coin flip.

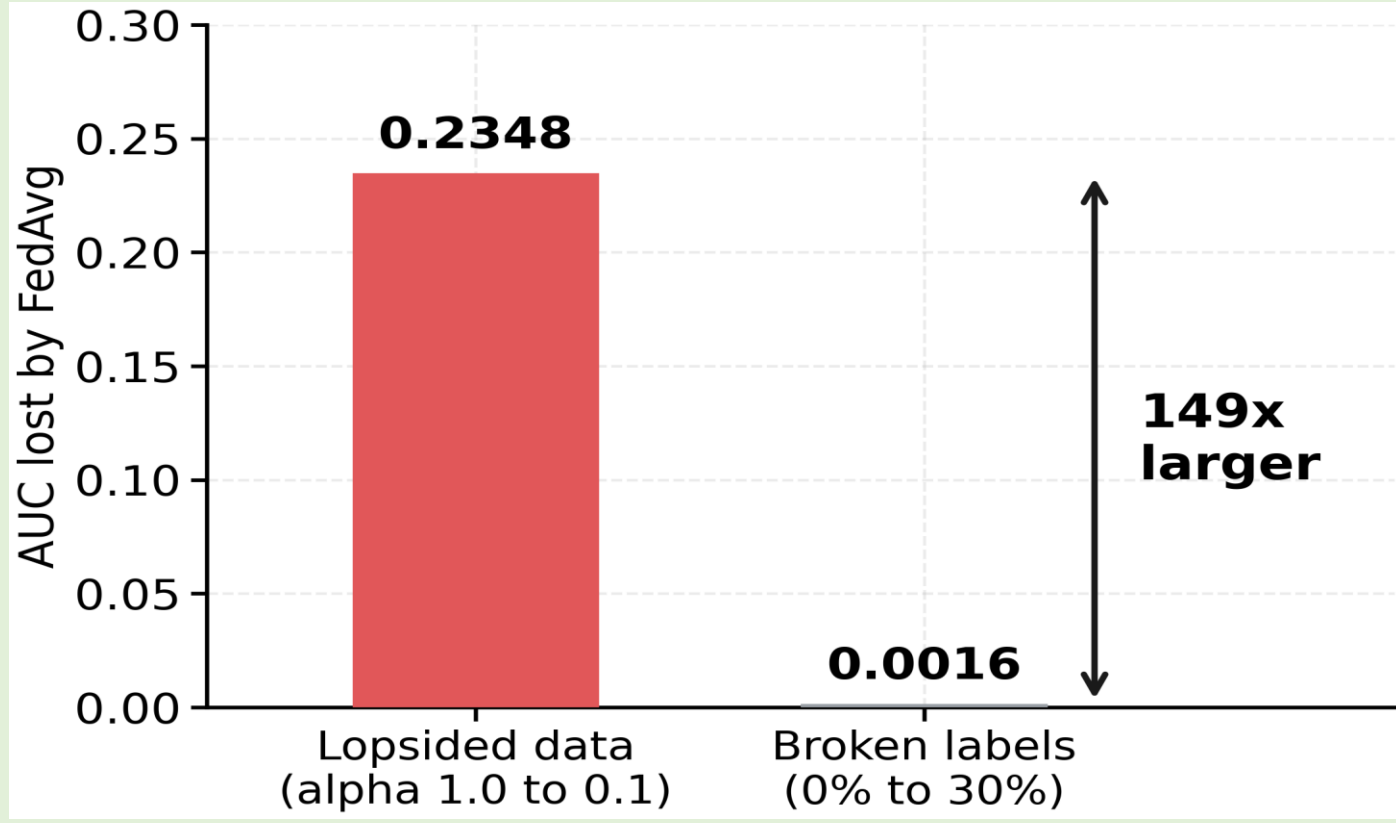
Code & Data

All code, the merged results and a script that re-checks every number on this poster are openly available.
github.com/kunalchevuri/federated-learning-health-robustness



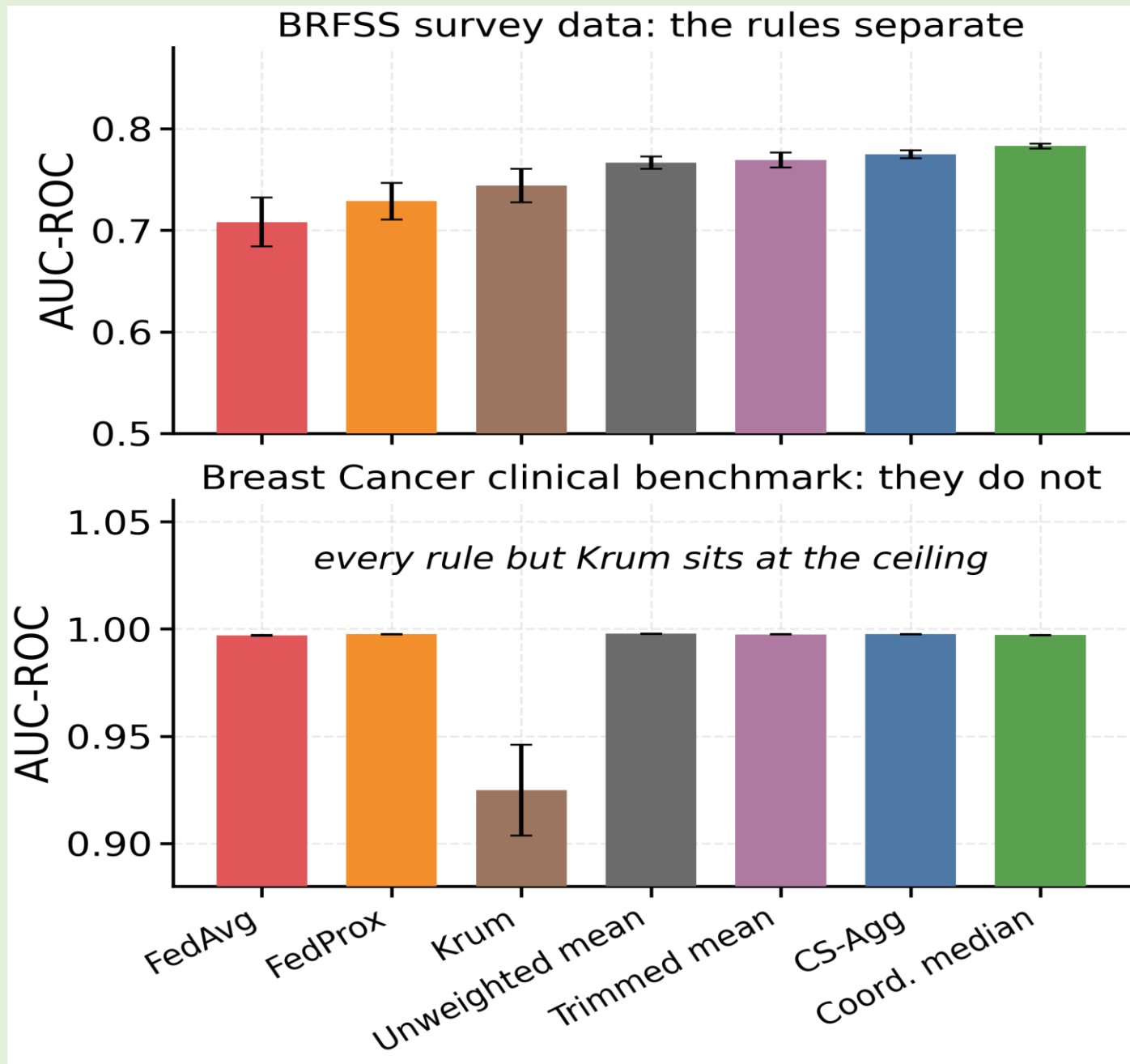
Results, continued

Lopsided data hurts far more than broken labels



- Moving from alpha = 1.0 to 0.1 costs FedAvg 0.2348 AUC, while 0% to 30% label noise costs it 0.0016.
- That 149x gap narrows to about 14x when sites track twelve health variables instead of four, but it never closes.

A clean benchmark would have hidden all of this



- On Breast Cancer every rule but Krum reaches 0.997 AUC or better, where no difference between them is visible.
- Messy, self-reported population health data is where these rules actually separate.

Future Directions

- Test adaptive Byzantine attackers, not only the label noise studied here.
- Replace the simulated Dirichlet split with data partitioned across real institutions.
- Check whether the weighting effect persists for deeper models and larger numbers of sites.

Conclusions

- Most of the advantage robust aggregators hold over FedAvg comes from how they weight sites, not from the robust statistic they are named for.
- Any robustness evaluation run on size-skewed data should include an unweighted-mean control.
- The control costs one line of code, and without it a reported gain cannot be credited to robustness.
- A rule's reputation is not evidence: Krum is widely cited and failed here, while coordinate-wise median is simple and held.

Acknowledgements

- Survey data collected and published by the Centers for Disease Control and Prevention (BRFSS 2023).
- Carried out in the Computational Healthcare and Biotechnology Lab, University of North Texas.